

v225 BEYOND-NMI differentiable- σ end-to-end model: learnable σ via softplus + differentiable Gaussian filtering. Network learns σ near {1.5, 2.4, 3.6, 4.9} ($\approx$ init). Pooled OOF AUC=0.605 — does NOT match fixed- σ logistic (0.708 OOF, 0.815 in-sample). ****Fixed- σ is already at the optimum**** — physics-grounded scales are intrinsically determined.

